Discrimination and Male-Female Earnings Differentials

EC306- Wages and Employment

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Goals of This Section

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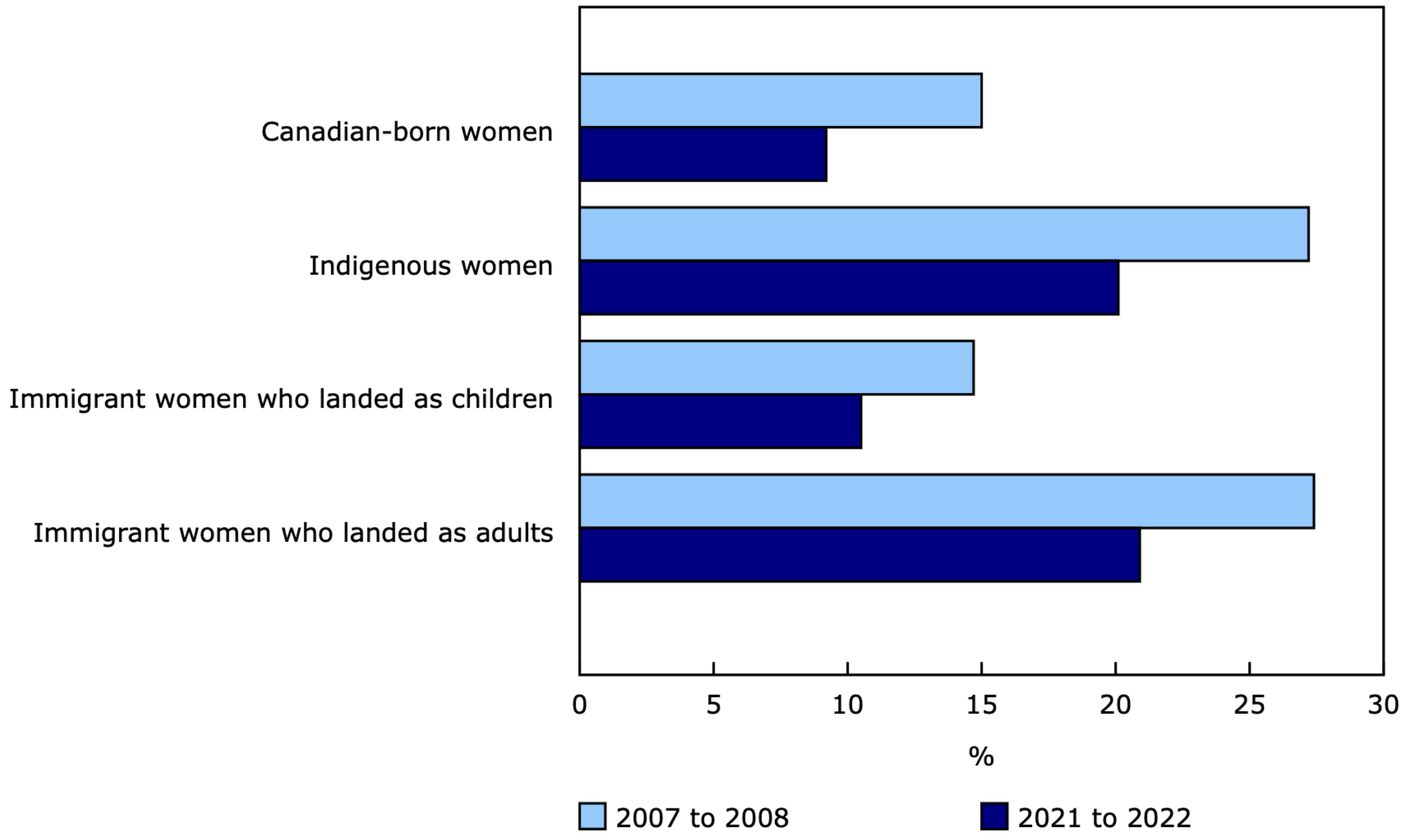
- Discuss trends in educational attainment in Canada
- Introduce Human Capital Theory of education investment
- Introduce Signalling Model of education investment
- Discuss empirical methods to estimate returns to schooling
- Review empirical evidence on returns to schooling
- Discuss job training

Introduction

Male and Female Earnings

Male and Female Earnings

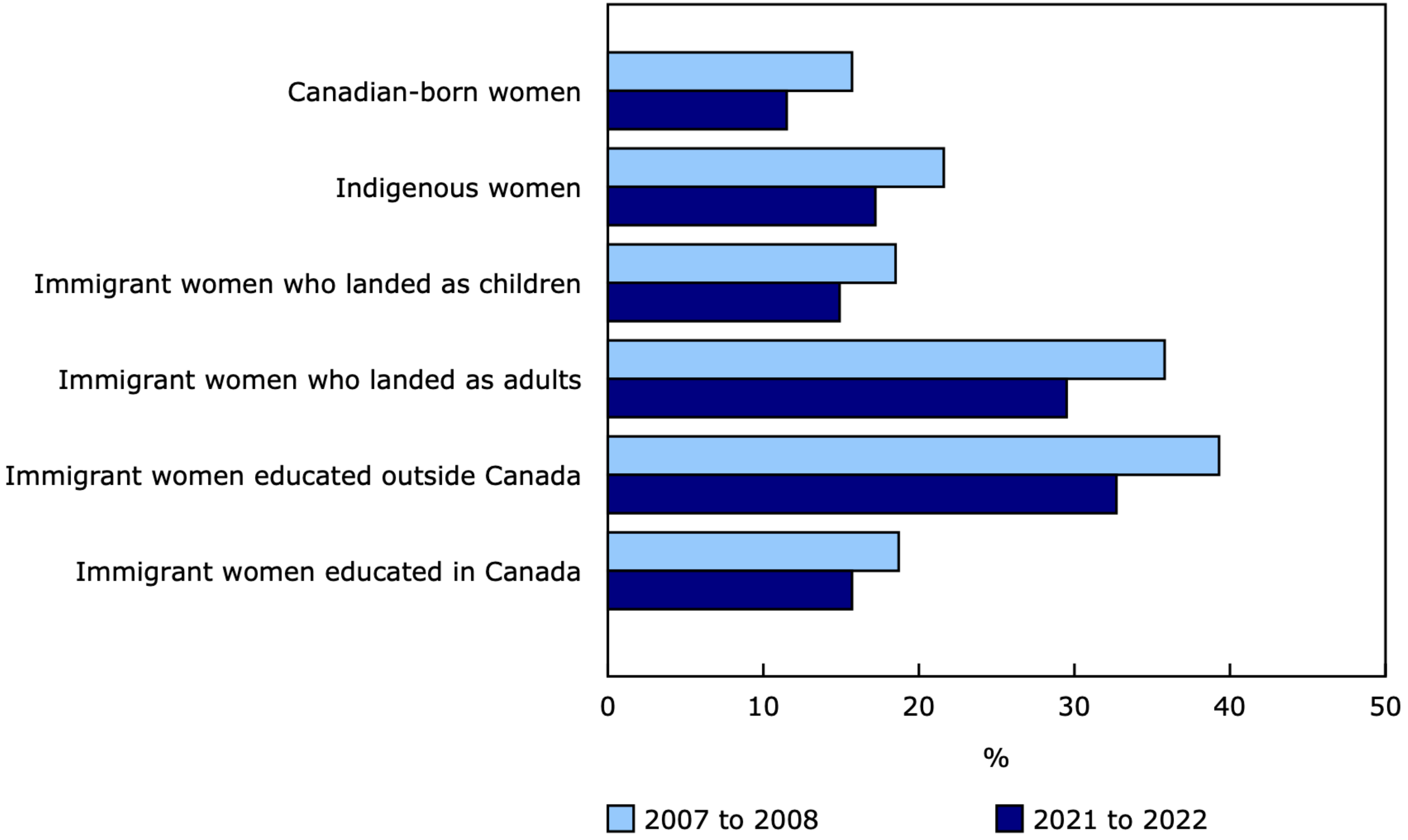
Gap in average hourly wage relative to Canadian-born men, by group of women, 2007 to 2008 and 2021 to 2022



Note(s): "Canadian-born" refers to non-Indigenous workers born in Canada.
Source(s): Labour Force Survey (3701), March and September monthly files, 2007, 2008, 2021, and 2022.

Male and Female Earnings

Gap in average hourly wage relative to Canadian-born men among workers holding a bachelor's degree or higher, 2007 to 2008 and 2021 to 2022



Note(s): "Canadian-born" refers to non-Indigenous workers born in Canada.
Source(s): Labour Force Survey (3701), March and September monthly files, 2007, 2008, 2021, and 2022.

Male and Female Earnings

- Clearly men and women earn different amounts on average
- There are two broad explanations for this gap:
  1. Differences in characteristics (education, experience, occupation, industry, hours worked, etc.)
  2. Differences in the returns to those characteristics (discrimination)
- Our models so far have not left room for discrimination
  - We treated all workers as the same
  - So there should not be any differences in returns

Male and Female Earnings

- In this section, we explore economic theories of discrimination
  - Can come from employers, customers, co-workers
  - Can be statistical or taste-based
- It is difficult empirically to estimate discrimination
  - Need to control for all relevant characteristics
  - Need to separate out differences in characteristics from differences in returns
- There are methods that can help us estimate discrimination
  - Oaxaca-Blinder decomposition
  - Audit studies
  - Experiments

Theories of Discrimination

Demand-Side Theories

- Demand theories focus on the behavior of employers in demanding labor
- Can happen for various reasons:
  - Pure prejudice
  - Customer preference to buy from male workers
  - Male aversion to working with women
- These all cause a fall in the demand for women relative to men in the labour market
  - Reduces wages and employment of female workers

Taste-Based Discrimination

- Theory of taste-based discrimination pioneered by Becker (1957)
- Imagine that firms produce according to the production function:

$$y = f(N_m + N_w)$$

- With N_m indicating the number of male workers, and N_f indicating the number of female workers
- So in a competitive market their profits are

$$\pi = p \cdot f(N_m + N_w) - w_m N_m - w_w N_w$$

- With w_m indicating the wage of male workers, and w_f indicating the wage of female workers

Taste-Based Discrimination

- Firms do not maximize profit strictly, but instead maximize a utility function

$$U = \pi - d \cdot N_w$$

- Utility is partly a function of profit
- But also includes d, which measures the degree of discrimination
  - if $d = 0$, no discrimination and maximizing utility is the same as maximizing profit
  - if $d = 0$ firms get a disutility from hiring women

Taste-Based Discrimination

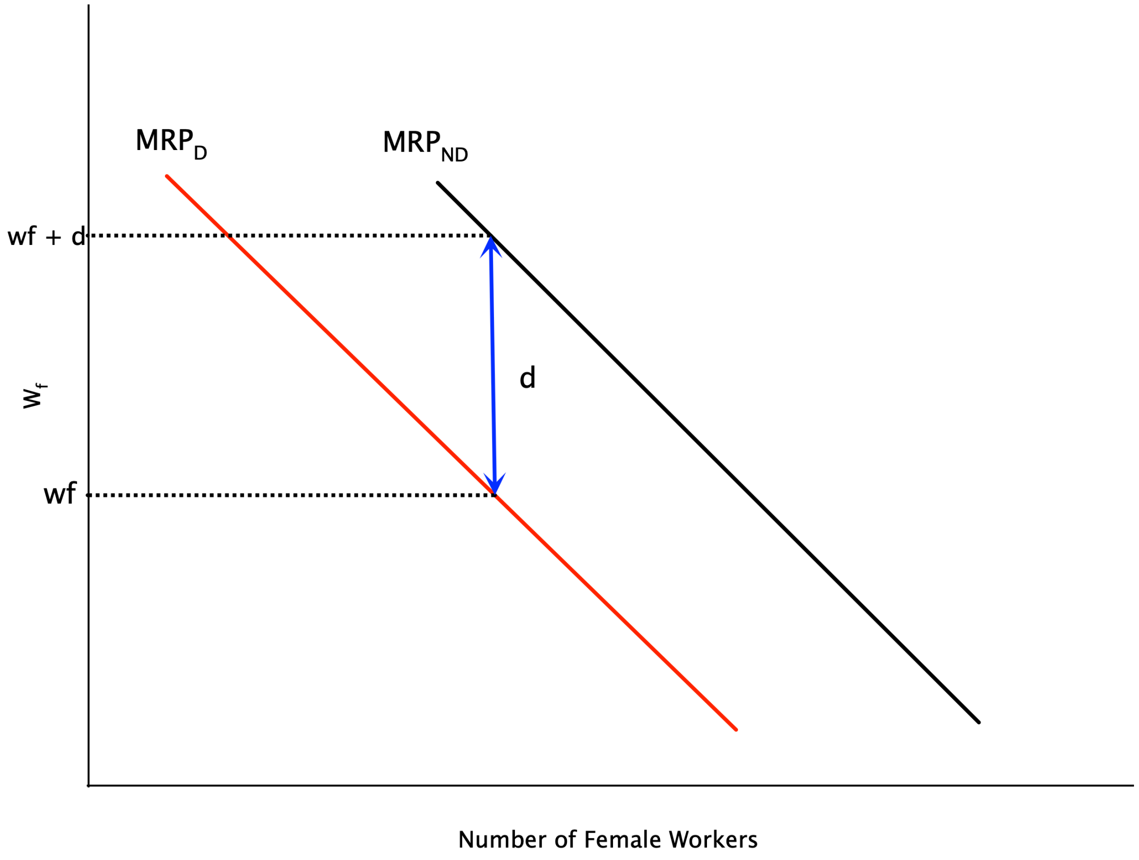
- The first order conditions for utility maximization are:

$$p \cdot MP_{N_m} = w_m$$

$$p \cdot MP_{N_w} = w_w + d$$

- For both men and women, firms hire up to the point where the value of the marginal product equals the cost of hiring
  - But for women, there is an additional cost d due to discrimination
- If $w_m = w_f$, and men and women are equally productive, this implies the firm hires less women
  - Discrimination drives a wedge between wages and the MRP

Taste-Based Discrimination

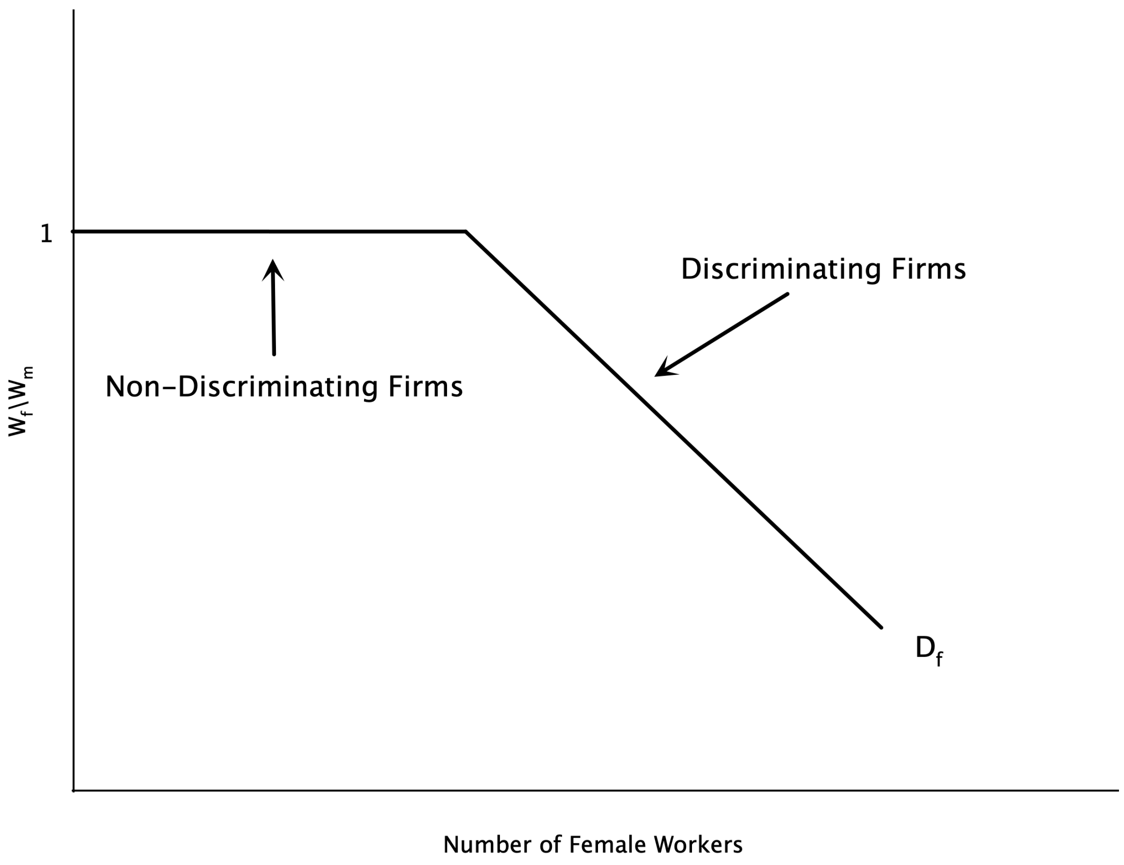


- Graph plots labour demand for women
- Discrimination shifts demand curve left
- Without discrimination, firm hires where $w_f = MRP_{ND}$
  - ND = non-discriminatory
- With discrimination, firm hires where $w_f + d = MRP_D$
  - Or equivalently $w_f = MRP_D - d$
- Implies a lower level of female labour

Taste-Based Discrimination

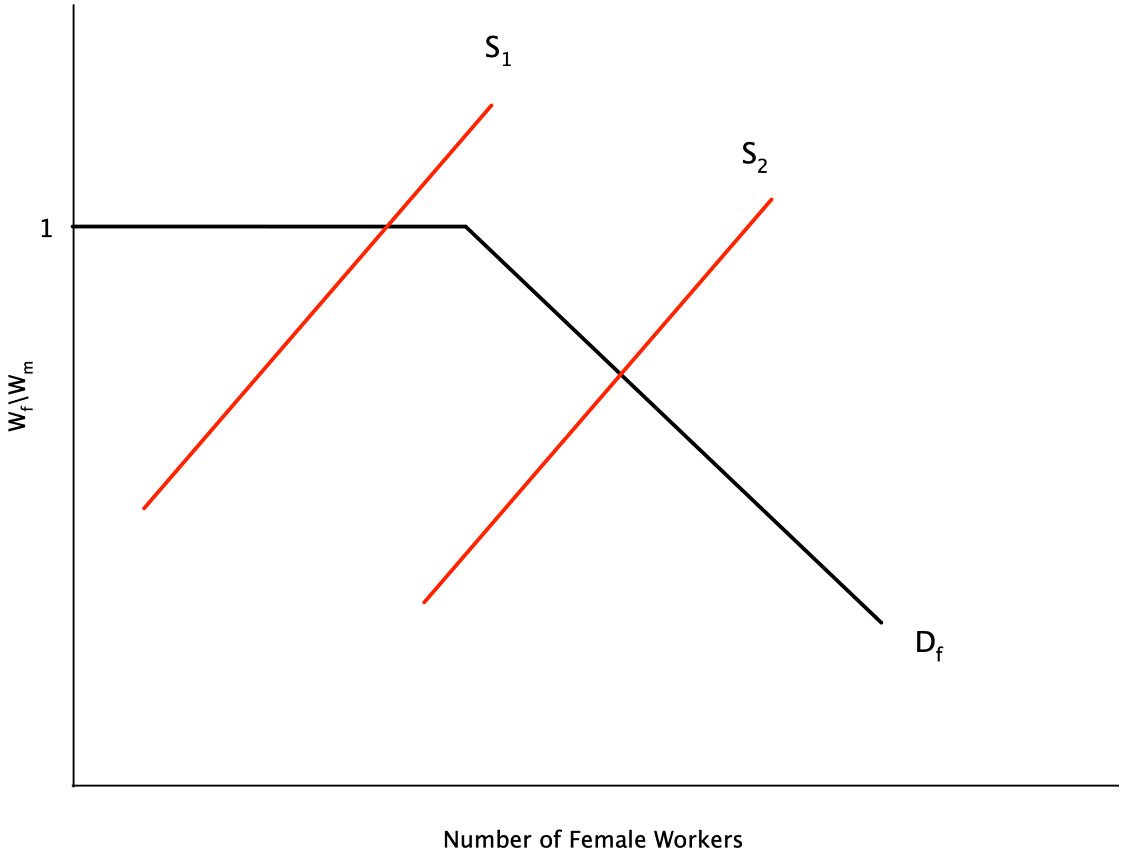
- Above analysis is for a single employer
- Employers discriminate along a spectrum
  - Some have a high d, some have a low or no d
- Suppose women are able to seek out the non-discriminating employers
- If we order the firms from lowest to highest by the amount of discrimination
  - We can get the market demand curve by summing the individual demand curves horizontally
- This demand curve is plotted on the next slide

Taste-Based Discrimination



- Graph plots demand for female labour in the market
- On the vertical axis is the ration of female to male wages
- For non-discriminating firms, this ratio is 1
  - A certain set of firms will hire women at this ratio
  - This is the flat portion of the demand curve
- As the supply of women grows, they must work at more discriminating firms
  - These firms will only hire when the wage ratio falls

Taste-Based Discrimination



- If the supply of female labour is small enough, they can work at non-discriminating firms
  - Wage ratio is 1
- With a larger supply, some will have to work at discriminating firms
  - Wage ratio falls below 1
  - This is where the demand curve slopes downwards
  - Wage gap emerges

Statistical Discrimination

- Taste-based discrimination relies on employers having a “taste” for discrimination
  - Overt prejudice
- Some discrimination may happen because of imperfect information
  - Employers do not know the productivity of workers perfectly
- Instead they use signals to infer worker quality
- With **statistical discrimination** firms use group characteristics to infer individual productivity
  - If one group has lower average productivity, firms may pay them less on average

Statistical Discrimination

Theory

- Assume that the abilities of male and female workers are distributed with means $\bar{a}_m$ and $\bar{a}_f$
- Suppose the firm has access to a signal of ability x for each person
- Actual ability is not known with certainty, but expected ability is

$$a(x) = \bar{a} + \alpha(x - \bar{a})$$

- where α is the reliability of the signal
  - If $\alpha = 0$, the signal provides no information beyond the average
  - If $\alpha = 1$, the signal is perfectly informative
- Workers are paid a wage equal to their marginal revenue product

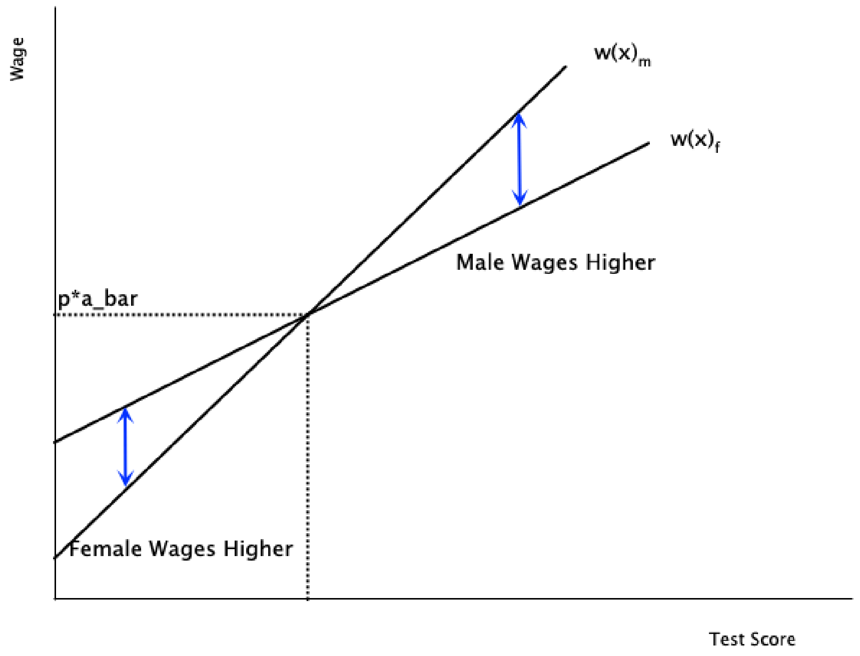
Statistical Discrimination

- Paying workers their marginal revenue product implies

$$w(x) = pa(x) = p\bar{a} + p\alpha(x - \bar{a})$$

- To analyze discrimination, assume $\bar{a}_m = \bar{a}_f$
  - Men and women are equally productive on average
- Assume the signal is more reliable for men than for women
  - If $\alpha_m > \alpha_f$
- Differences arise between workers who score the same on the signal
  - If a man and woman score above average $(x - \bar{a} > 0)$, then $w(x)_m > w(x)_f$
  - If a man and woman score below average $(x - \bar{a} < 0)$, then $w(x)_m < w(x)_f$

Statistical Discrimination



- In this graph we have wages as a function of the signal
- The male line is steeper because the signal is more reliable ($\alpha_m > \alpha_f$)
- For test scores above average, men are paid more
- For test scores below average, women are paid more
- At the average, they are paid the same

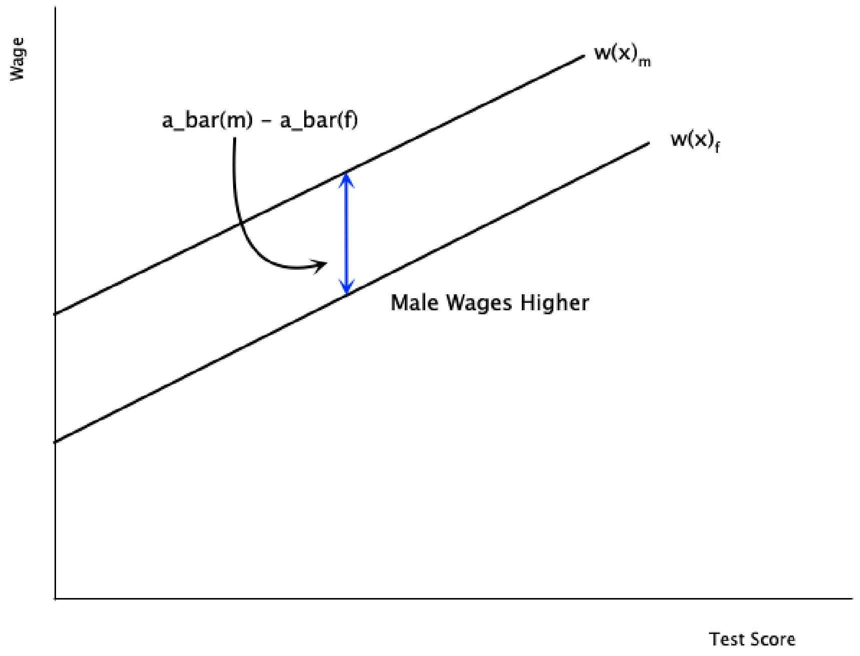
Statistical Discrimination

- There are no underlying differences between males and females
  - Differences in wages arise because of differences in the quality of the signal
  - Firms place more weight on average ability for female workers
  - Firms place more weight on the actual signal for male workers
- At low levels of the signal, women are paid more than men
- But where does the assumption $\alpha_m > \alpha_f$ come from
  - Possibly related to prejudicial attitudes of firms
  - Some believe standardized tests favour males

Statistical Discrimination

- There may also be differences in $\bar{a}_m$ and $\bar{a}_f$
  - Possibly due to pre-market discrimination leading to lower human capital among women
- In this case, males are paid more at every signal level
- Wage differences do not necessarily arise from overt discrimination
  - They may originate in differences in information signals
  - But it is difficult to explain why such differences in signal quality exist

Statistical Discrimination



- In this graph the quality of the signal is the same
- Slopes of the lines are the same
- But there are differences in average productivity
  - Male line lies above female line
- Men will be paid more because of the average difference